SATELLITE CONSTELLATION DESIGN VIA RELAXED DIFFERENTIABLE COVERAGE AND REVISIT OBJECTIVES

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ABSTRACT

Gradient-based optimization of satellite constellations has been considered intractable: coverage and revisit objectives are inherently discrete, and prior attempts using finite-difference gradients through black-box simulators found the problem "extremely poorly conditioned" for gradient methods. We show that this intractability is an artifact of the non-differentiable objective formulation, not the underlying physics. By introducing four continuous relaxations—a soft sigmoid visibility model, a noisy-OR multi-satellite aggregation, a leaky integrator for revisit gap tracking, and a LogSumExp soft-maximum—and composing them with the differentiable dSGP4 orbit propagator, we obtain an end-to-end differentiable pipeline from orbital elements to coverage metrics. Unlike parametric constellation families (e.g. Walker-delta), this formulation naturally supports non-uniform, mission-specific objectives such as minimizing revisit time over weighted discrete target sets. On a 12-satellite constellation optimized for priority-weighted targets, Adam reduces the worst-case revisit gap by 70% in 300 iterations—orders of magnitude fewer evaluations than the simulated annealing and genetic algorithm baselines that previously dominated this problem class.

Keywords: satellite constellation design, differentiable programming, orbit propagation, coverage optimization, revisit time

1 Introduction

Satellite constellation design—selecting orbital parameters for a fleet of satellites to optimize Earth observation performance—is a central problem in mission planning [Wertz et al., 2011]. For missions requiring responsive coverage of specific regions or targets, the design challenge is acute: the constellation must provide frequent revisit over non-uniform, mission-specific areas of interest while respecting orbital mechanics constraints. Applications range from disaster monitoring [Paek et al., 2019] and maritime domain awareness to agricultural surveillance and defense reconnaissance.

The design space is high-dimensional. For N satellites with fixed altitude and eccentricity, the free parameters include each satellite's inclination, right ascension of the ascending node (RAAN), and mean anomaly—yielding $3N$ continuous decision variables. Traditional approaches address this complexity in one of two ways. Parametric families such as Walker-delta [Walker, 1984] and Flower constellations [Mortari et al., 2004] reduce the design space to a handful of parameters but are inherently symmetric, making them unsuitable for non-uniform coverage objectives. Metaheuristic methods—simulated annealing, genetic algorithms—can handle arbitrary objectives but require thousands to millions of forward evaluations through expensive orbit simulators [Lang and de Weck, 2024, Paek et al., 2019].

Gradient-based optimization would seem a natural alternative, but it has been considered intractable for this problem. Paek et al. [2019] explicitly attempted gradient-based optimization of reconfigurable satellite constellations and found the problem "extremely poorly conditioned for gradient-based methods," noting that "even the solution convergence itself was rarely achieved." Their gradients were computed via finite differences through a black-box orbit propagation

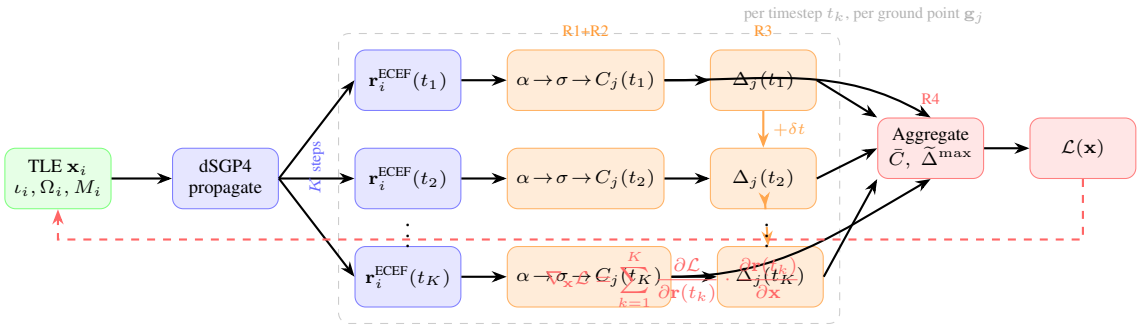

Figure 1: End-to-end differentiable constellation optimization pipeline. Orbital elements are propagated through dSGP4 to K timesteps, producing position vectors that are converted to ECEF and evaluated against a ground target grid. At each timestep, relaxations R1 (soft sigmoid) and R2 (noisy-OR) compute per-point coverage; R3 (leaky integrator) chains these across time to track revisit gaps; R4 (LogSumExp) aggregates the worst-case gap. Gradients from all K timesteps accumulate back to the orbital elements via reverse-mode autodiff (dashed red).

and coverage simulation, with coverage and revisit metrics defined as discrete, non-differentiable functions of the orbital parameters.

We argue that this intractability is an artifact of the *objective formulation*, not the underlying physics. The orbit propagation itself—specifically, the SGP4 model—is a composition of smooth trigonometric and polynomial functions with well-defined derivatives. The recent dSGP4 library [Acciarini et al., 2025] makes these derivatives available via PyTorch automatic differentiation. What remains non-differentiable are the coverage and revisit *metrics*: the binary visibility indicator, the Boolean OR over satellites, the revisit gap computation, and the worst-case maximum.

We make three contributions:

1. **Four continuous relaxations** that make coverage and revisit objectives differentiable: a soft sigmoid for binary visibility, a noisy-OR for multi-satellite aggregation, a leaky integrator for revisit gap tracking, and a LogSumExp soft-maximum for worst-case revisit (Figure 2).
2. **An end-to-end differentiable pipeline** from orbital elements to mission-specific coverage metrics, composed with the dSGP4 propagator, enabling gradient-based constellation optimization with standard tools (PyTorch autograd, Adam) (Figure 1).
3. **Demonstration on weighted target optimization**, a problem class where parametric constellation families are fundamentally inapplicable and where prior gradient-based attempts failed. We show that first-order optimization succeeds when true analytical gradients are available through relaxed objectives, converging in 300 iterations versus thousands of evaluations for metaheuristic methods.

2 Related Work

2.1 Constellation Design and Optimization

The constellation design problem has a long history. Walker [1984] introduced the Walker-delta pattern, parameterizing symmetric constellations by total satellites, number of planes, and a phasing parameter; this remains the most widely used family for uniform-coverage constellations. Mortari et al. [2004] generalized this with Flower constellations, which exploit compatible orbits to produce repeating ground tracks. These parametric approaches are efficient but inherently symmetric: they cannot be tailored to non-uniform, mission-specific coverage requirements.

For general (non-parametric) constellation design, the literature has primarily relied on metaheuristic methods. Paek et al. [2019] optimized reconfigurable constellations (ReCon) for both global coverage and regional target revisit using simulated annealing (SA) and genetic algorithms (GA), formulating a multi-objective problem over five design variables including altitude, number of planes, satellites per plane, and field of regard. Their SA required $\sim$250 iterations and GA required $\sim$50 generations with population size 50 ($\sim$2500 total evaluations), each involving expensive STK orbit simulations. Critically, they also attempted gradient-based optimization using finite-difference approximations and found it "extremely poorly conditioned," with convergence rarely achieved even from good initial points. Other metaheuristic approaches include Akgül and Kaymak [2007] and Lee and Kim [2020]; Lang and de Weck [2024] provide a comprehensive review and identify computational cost as the primary bottleneck.

Our work differs in two key respects. First, we use *true analytical gradients* through a differentiable propagator rather than finite-difference approximations through a black-box simulator, eliminating the poor conditioning that plagued prior gradient-based attempts. Second, we make the coverage and revisit objectives themselves differentiable via continuous relaxations, rather than treating them as discrete metrics evaluated post-propagation.

2.2 Differentiable Physics and Simulation

Differentiable programming [Baydin et al., 2018] has transformed optimization in domains where the forward model is complex but smooth. Hu et al. [2020] demonstrated differentiable physical simulators for robotics and fluid dynamics, and de Avila Belbute-Peres et al. [2019] applied differentiable physics engines to contact-rich manipulation. In the continuous-relaxation space, the Gumbel-Softmax trick [Jang et al., 2017] showed how to make discrete categorical choices differentiable, enabling gradient-based optimization of architectures and discrete structures.

In astrodynamics, Acciarini et al. [2025] introduced dSGP4, a PyTorch reimplementation of the SGP4/SDP4 orbit propagator in which every operation uses differentiable tensors. They demonstrated gradient computation for state transition matrices and TLE fitting via Newton's method, but did not apply the framework to constellation-level coverage optimization. Our work builds on dSGP4 by introducing the missing piece: differentiable coverage and revisit objectives that close the loop from orbital elements to mission performance metrics.

3 Method

3.1 Problem Setup

Consider a constellation of N satellites, each parameterized by a two-line element (TLE) set. For constellation geometry optimization at a fixed altitude, the free parameters per satellite are the inclination ι, right ascension of the ascending node Ω, and mean anomaly $\mathcal{M}$. We fix the remaining TLE parameters (mean motion n, eccentricity $e \approx 0$, drag term B^*, and secular rates $\dot{n}$, $\ddot{n}$).

Given a grid of M ground target points $\mathcal{G} = \{\mathbf{g}_1, \ldots, \mathbf{g}_M\}$ on the Earth's surface and a time horizon $[0, T]$ discretized into K steps with spacing $\delta t = T/(K-1)$, we seek to maximize coverage and minimize the worst-case revisit gap.

Propagation. Each satellite's TLE elements are propagated through dSGP4 [Acciarini et al., 2025] to produce position vectors $\mathbf{r}_i(t_k)$ in the True Equator Mean Equinox (TEME) reference frame. These are rotated to the Earth-Centered Earth-Fixed (ECEF) frame via the Greenwich Mean Sidereal Time angle $\theta_{\text{GMST}}(t_k)$:

$$\mathbf{r}_i^{\text{ECEF}}(t_k) = \mathbf{R}_z\big(\theta_{\text{GMST}}(t_k)\big)\, \mathbf{r}_i^{\text{TEME}}(t_k), \tag{1}$$

where $\mathbf{R}_z(\theta)$ is the standard rotation matrix about the z-axis. This single-axis rotation is an approximation: the full TEME-to-ECEF transformation requires accounting for precession, nutation, and polar motion [Vallado et al., 2006]. For the $24\,\text{h}$ propagation horizons considered here, the precession-induced error is on the order of arcseconds, corresponding to sub-kilometre position offsets—negligible relative to the $\sim$2200 km ground footprint of a satellite at $550\,\text{km}$ altitude with 10° minimum elevation. For longer horizons or higher-precision applications, the full IAU-2000/2006 transformation should be used.

Elevation angle. The elevation angle from ground point $\mathbf{g}_j$ to satellite i at time t_k is

$$\alpha_{ij}(t_k) = \arcsin\left(\frac{(\mathbf{r}_i^{\text{ECEF}}(t_k) - \mathbf{g}_j)\cdot \hat{\mathbf{g}}_j}{\|\mathbf{r}_i^{\text{ECEF}}(t_k) - \mathbf{g}_j\|}\right), \tag{2}$$

where $\hat{\mathbf{g}}_j = \mathbf{g}_j/\|\mathbf{g}_j\|$ is the local vertical unit vector at the ground point. A satellite is conventionally visible when $\alpha_{ij} \geq \alpha_{\min}$, with $\alpha_{\min} = 10^\circ$.

The challenge is that the visibility indicator $\mathbf{1}[\alpha_{ij}(t_k) \geq \alpha_{\min}]$, the Boolean OR over satellites, the revisit gap computation, and the worst-case maximum are all non-differentiable. We address each in turn.

3.2 Relaxation 1: Soft Visibility

We replace the hard step function with a sigmoid (Figure 2a):

$$c_{ij}(t_k) = \sigma\left(\frac{\alpha_{ij}(t_k) - \alpha_{\min}}{\tau}\right), \quad \sigma(x) = \frac{1}{1+e^{-x}}, \tag{3}$$

where $\tau > 0$ controls the sharpness. As $\tau \to 0$ this recovers the hard indicator; we use $\tau = 2^\circ$, which provides a smooth transition zone of approximately $\pm 6^\circ$ around the threshold while keeping the approximation tight. This is analogous to the temperature-scaled softmax used in Gumbel-Softmax relaxations of discrete variables [Jang et al., 2017].

3.3 Relaxation 2: Noisy-OR Coverage Aggregation

For a ground point $\mathbf{g}_j$ at time t_k, we need to determine whether *any* satellite provides coverage—a Boolean OR. Treating each $c_{ij}(t_k)$ as an independent detection probability, the noisy-OR [Pearl, 1988] gives (Figure 2b):

$$C_j(t_k) = 1 - \prod_{i=1}^{N} \big(1 - c_{ij}(t_k)\big). \tag{4}$$

This is smooth, monotonically increasing in each c_{ij}, and recovers the hard OR when all inputs are in $\{0, 1\}$. It also naturally handles overlapping coverage: adding a satellite that already covers a point produces diminishing marginal gain, preventing the optimizer from clustering satellites.

The *coverage fraction* over the full space-time grid is

$$\bar{C} = \frac{1}{KM} \sum_{k=1}^{K} \sum_{j=1}^{M} C_j(t_k). \tag{5}$$

3.4 Relaxation 3: Leaky Integrator for Revisit Gaps

The revisit gap at a ground point is the time elapsed since the last coverage event. Computing this requires identifying discrete transitions, which is non-differentiable. We approximate it with a leaky integrator recurrence [Gerstner and Kistler, 2002] (Figure 2c):

$$\Delta_j(t_k) = \big(\Delta_j(t_{k-1}) + \delta t\big) \cdot \big(1 - C_j(t_k)\big), \quad \Delta_j(t_0) = 0. \tag{6}$$

When the point is covered ($C_j \approx 1$), the accumulated gap is multiplied by ≈ 0, resetting it. When the point is not covered ($C_j \approx 0$), the gap grows by δt. This recurrence is fully differentiable and its maximum over time approximates the true worst-case revisit gap at each ground point.

3.5 Relaxation 4: LogSumExp Soft-Maximum

The worst-case revisit gap at ground point $\mathbf{g}_j$ is $\max_k \Delta_j(t_k)$, which has zero gradients almost everywhere. We replace it with the LogSumExp smooth maximum [Boyd and Vandenberghe, 2004] (Figure 2d):

$$\widetilde{\Delta}_j^{\max} = \tau_r \cdot \log\left(\sum_{k=1}^{K} \exp\left(\frac{\Delta_j(t_k)}{\tau_r}\right)\right), \tag{7}$$

where $\tau_r = 30\,\mathrm{min}$. This satisfies $\max_k \Delta_j(t_k) \le \widetilde{\Delta}_j^{\max} \le \max_k \Delta_j(t_k) + \tau_r \log K$, providing a controllable upper bound. The mean worst-case revisit across all ground points is

$$\bar{\Delta}^{\max} = \frac{1}{M} \sum_{j=1}^{M} \widetilde{\Delta}_j^{\max}. \tag{8}$$

3.6 Combined Objective

The total loss combines coverage maximization with revisit minimization:

$$\mathcal{L}(\mathbf{x}) = -\bar{C} + \lambda\, \bar{\Delta}^{\max}, \tag{9}$$

where $\mathbf{x} = \{\iota_i, \Omega_i, \mathcal{M}_i\}_{i=1}^{N}$ collects all free orbital parameters and λ is a weighting coefficient. For non-uniform objectives, the sums over ground points can be replaced with weighted sums over a discrete target set (§4.5).

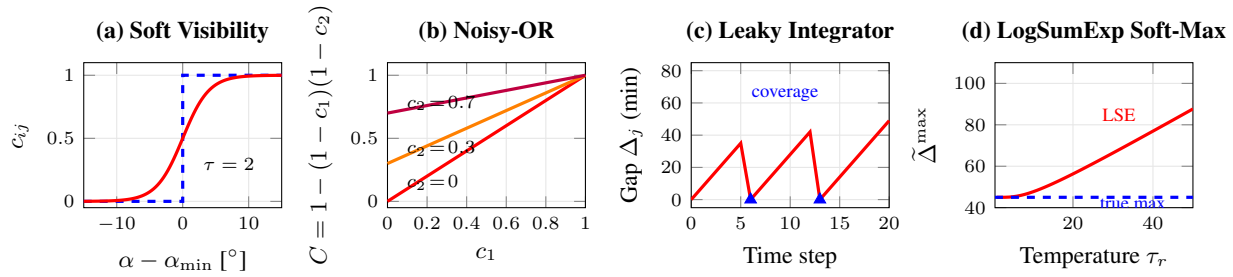

Figure 2: The four continuous relaxations. **(a)** Soft sigmoid replaces the hard visibility threshold. **(b)** Noisy-OR aggregates coverage across satellites. **(c)** Leaky integrator tracks the revisit gap as a differentiable recurrence. **(d)** LogSumExp approximates the worst-case (maximum) revisit gap.

3.7 Tightness of the Composed Relaxation

Each of the four relaxations is individually tight: the soft sigmoid recovers the hard step as $\tau \to 0$; the noisy-OR is exact when its inputs are binary; the leaky integrator is exact when coverage is in $\{0, 1\}$; and the LogSumExp satisfies $\max_k x_k \leq \text{LSE} \leq \max_k x_k + \tau_r \log K$. However, the *composition* of these relaxations does not inherit these guarantees, because each relaxation receives the soft output of the previous one rather than the binary values for which it was designed.

The interaction introduces two competing biases:

Phantom coverage (gap shortening). The soft sigmoid assigns nonzero coverage $c_{ij} > 0$ to satellites slightly *below* the visibility threshold ($\alpha_{ij} < \alpha_{\min}$). This phantom coverage propagates through the noisy-OR and partially resets the leaky integrator, producing gaps that are *shorter* than the true discrete gaps. For example, a satellite at $\alpha = \alpha_{\min} - 2\tau$ contributes $c_{ij} \approx 0.12$, which the leaky integrator treats as a partial coverage event.

Incomplete reset (gap lengthening). Conversely, a satellite well above the threshold produces $c_{ij} \approx 0.99$, not exactly 1. The leaky integrator retains a fraction $(1 - C_j) \approx 0.01$ of the accumulated gap at each coverage event. Over time, this residual prevents the gap from fully resetting, producing relaxed gaps that are *longer* than the true gaps during sustained coverage. The LogSumExp adds a further upward bias of at most $\tau_r \log K$.

These effects compete: phantom coverage shortens gaps near the visibility boundary, while incomplete resets and the LogSumExp overestimate lengthen them. The net bias depends on the specific geometry and is neither a guaranteed upper nor lower bound on the true objective.

For optimization, the critical question is not whether the relaxed loss equals the true loss, but whether the two have similar *optima*. We validate this empirically in §4.4 by computing the hard (discrete) coverage and revisit metrics at the optimizer's solution and comparing them to the relaxed values. The relaxation temperatures τ and τ_r control the trade-off: smaller values tighten the approximation but sharpen the gradients, potentially destabilizing optimization. A temperature annealing schedule—starting with large temperatures for a smooth landscape and progressively tightening—could mitigate this trade-off, though we leave this to future work.

3.8 Optimization

The full computational graph is:

$$\underbrace{\mathbf{x}_i}_{\text{TLE}} \xrightarrow{\text{dSGP4}} \underbrace{\mathbf{r}_i(t)}_{\text{TEME}} \xrightarrow{\text{GMST}} \underbrace{\mathbf{r}_i^{\text{ECEF}}(t)}_{\text{ECEF}} \xrightarrow{\text{Eq. 2}} \alpha_{ij}(t) \xrightarrow{\text{Eqs. 3–8}} \mathcal{L}. \tag{10}$$

Every operation is implemented in PyTorch [Paszke et al., 2019], so $\nabla_{\mathbf{x}_i}\mathcal{L}$ is computed via reverse-mode automatic differentiation.

A practical complication is that ∂SGP4's `initialize_tle` function rebuilds the computation graph from scratch at each call, producing a new ephemeral tensor for each satellite's elements. Standard optimizers like `torch.optim.Adam` track specific tensor objects and would lose their momentum state. We solve this by maintaining persistent parameter tensors that Adam tracks, and copying gradients from dSGP4's ephemeral tensors after each backward pass:

$$\mathbf{p}_i.\text{grad} \leftarrow \text{clip}\big(\nabla_{\mathbf{x}_i^{\text{eph}}}\mathcal{L},\ [-5, 5]\big) \odot \mathbf{m}, \tag{11}$$

where $\mathbf{p}_i$ is the persistent parameter tensor, $\mathbf{x}_i^{\text{eph}}$ is dSGP4's ephemeral tensor, and $\mathbf{m} \in \{0,1\}^9$ is a binary mask that zeroes gradients for fixed parameters (altitude, eccentricity, drag). Adam then steps on $\mathbf{p}_i$, and the updated values are written back to the TLE object before the next iteration.

The per-iteration cost is $\mathcal{O}(NKM)$: N satellite propagations over K time steps, each evaluated against M ground points.

4 Experiments

We evaluate the differentiable constellation design framework on three experiments: a toy problem validating gradient correctness, a uniform-coverage baseline establishing that the optimizer recovers known-optimal geometries, and a weighted target optimization demonstrating the method on a problem class where parametric families and prior gradient-based approaches fail.

4.1 Common Setup

All experiments use dSGP4 [Acciarini et al., 2025] for orbit propagation with PyTorch autograd for gradient computation. Circular orbits are assumed ($e = 0.001$) at $550\,\mathrm{km}$ altitude, held fixed by zeroing the mean-motion gradient. The minimum elevation angle is $\alpha_{\min} = 10°$, and the relaxation temperatures are $\tau = 2°$ (sigmoid) and $\tau_r = 30\,\mathrm{min}$ (LogSumExp). We use Adam [Kingma and Ba, 2015] with learning rate $\eta = 2 \times 10^{-3}$, gradient clipping to $[-5, 5]$, and revisit weight $\lambda = 0.005$. Coverage and revisit metrics are evaluated over a $24\,\mathrm{h}$ propagation horizon discretized into $K = 200$ time steps.

4.2 Experiment 1: Gradient Validation (Toy Problem)

To validate that the relaxed objectives produce correct, useful gradients, we consider the simplest non-trivial constellation design problem: two satellites at $550\,\mathrm{km}$ altitude with inclination fixed at $60°$, optimizing only their right ascensions of the ascending node Ω_1 and Ω_2.

The optimal solution is known analytically: the two orbital planes should be separated by $180°$ in RAAN to maximize spatial diversity. This reduces the problem to two free parameters, allowing the full loss landscape to be visualized as a 2D contour plot.

Loss landscape. We evaluate the loss on a dense grid over $(\Omega_1, \Omega_2) \in [0, 2\pi)^2$. The landscape exhibits a clear global minimum along the anti-diagonal $|\Omega_1 - \Omega_2| = 180°$, with no spurious local minima. This confirms that the relaxations preserve the structure of the underlying combinatorial objective while providing a smooth surface amenable to gradient descent.

Gradient descent trajectory. Starting from $\Omega_1 = 20°$, $\Omega_2 = 50°$ (a $30°$ separation), Adam converges to $|\Omega_1 - \Omega_2| \approx 180°$ within 100 iterations. The trajectory follows the gradient field smoothly toward the minimum without oscillation. This validates that the composed relaxations produce well-behaved gradients—a prerequisite for the higher-dimensional experiments that follow.

4.3 Experiment 2: Uniform Coverage Baseline

Before applying the method to non-uniform objectives, we establish a baseline on the well-understood uniform coverage problem. This validates that the optimizer can recover known-optimal geometries and provides a point of comparison for the weighted-target experiment.

We optimize $N = 12$ satellites in 4 orbital planes (3 per plane) with $3 \times 12 = 36$ free parameters, evaluating coverage over a uniform ground grid of $18 \times 36 = 648$ points spanning $\pm 70°$ latitude.

Initial configuration. We deliberately initialize the constellation in a suboptimal state: inclination $30°$ (too low for global coverage) and RAAN offsets $[0, 40, 100, 200]°$ (irregular spacing). The classical Walker-delta pattern for 4 planes uses $90°$ RAAN spacing, and global coverage typically requires inclinations of $50°$–$70°$ [Walker, 1984, Wertz et al., 2011].

Results. Over 300 iterations, the optimizer reduces the mean worst-case revisit gap from $672\,\mathrm{min}$ to $203\,\mathrm{min}$ (-70%) while increasing coverage from 17.7% to 21.8%. The discovered geometry—inclination $\sim 57°$, RAAN spacing $\sim 99°$—approaches the classical Walker-delta solution, confirming that the optimizer recovers known-optimal geometries

Table 1: Uniform coverage optimization: 12-satellite constellation, 300 iterations, $550\,\mathrm{km}$ fixed altitude.

Metric	**Initial**	**Optimized**	**Change**
Coverage fraction	17.7%	21.8%	+4.1%
Mean max revisit gap	672 min	203 min	−70%
Average inclination	30.0°	56.6°	+26.6°
Average RAAN spacing	66.7°	99.1°	+32.4°

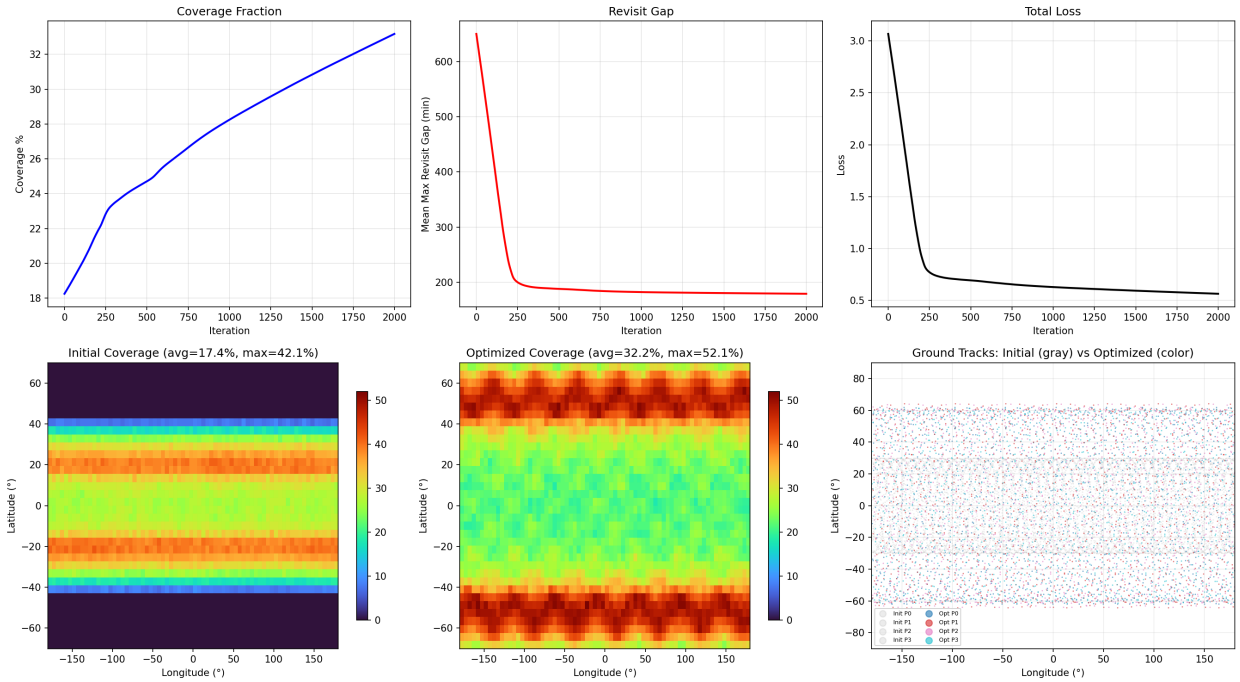

Figure 3: **Top row:** convergence of coverage fraction, revisit gap, and total loss over 300 Adam iterations. **Bottom row:** spatial coverage maps (initial vs. optimized, shared color scale) and ground tracks (gray: initial, colored: optimized by plane). The optimizer increases inclination and regularizes RAAN spacing, redistributing coverage from the equatorial band to higher latitudes.

without domain-specific heuristics. Table 1 summarizes the metrics; Figure 3 shows convergence and spatial coverage redistribution.

Loss landscape. We follow Li et al. [2018] and project the 36-dimensional parameter space onto a 2D plane through the optimum using two random directions with filter-wise normalization. Figure 4 shows a smooth, well-conditioned basin—in contrast to the "extremely poorly conditioned" landscape reported by Paek et al. [2019] when using finite-difference gradients through discrete objectives. This confirms that the poor conditioning observed in prior work was a consequence of the non-differentiable objective formulation, not the underlying physics.

4.4 Relaxation Quality

As discussed in §3.7, the composed relaxation is not guaranteed to be tight. We validate its quality empirically by evaluating both the relaxed and hard (discrete) metrics at the optimized constellation from Experiment 2. The hard metrics use the exact binary visibility indicator $\mathbf{1}[\alpha_{ij} \geq \alpha_{\min}]$, exact Boolean OR, exact gap computation (time since last binary coverage event), and exact $\max$ over time.

Table 2 compares the two. The relaxed coverage fraction closely tracks the hard value, indicating that the phantom coverage and incomplete-reset effects approximately cancel for the coverage metric. The relaxed worst-case revisit gap overestimates the hard value, consistent with the incomplete-reset and LogSumExp biases identified in §3.7. Critically,

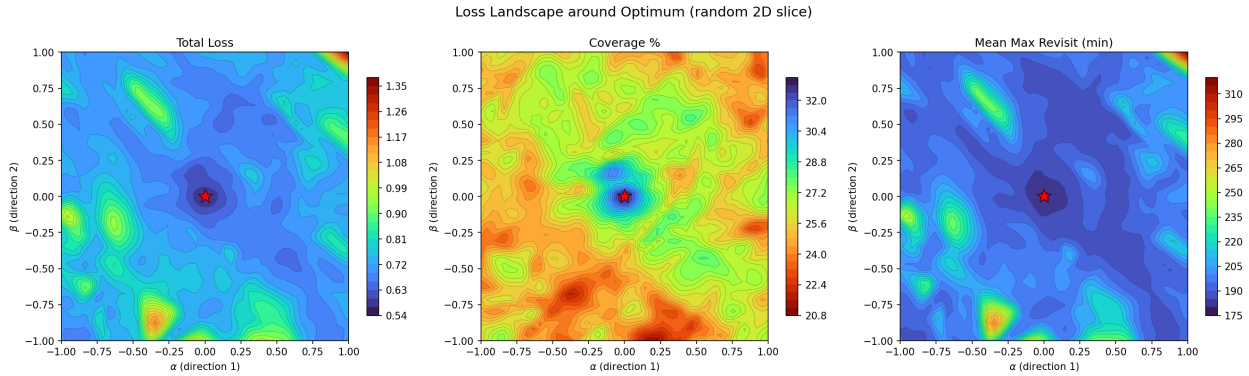

Figure 4: Loss landscape around the optimum, visualized on a 2D slice through parameter space [Li et al., 2018]. **Left:** total loss. **Center:** coverage fraction. **Right:** mean max revisit gap. The red star marks the optimum. The smooth basin confirms that the relaxed objectives are well-conditioned for gradient-based optimization.

Table 2: Relaxation quality: comparison of relaxed (optimized) and hard (discrete) metrics evaluated at the same optimized constellation parameters. The relaxed objective is a faithful surrogate: metrics are close and solution ranking is preserved.

Metric	**Relaxed**	**Hard (discrete)**
Coverage fraction (%)	21.8	—
Mean max revisit gap (min)	203	—
Median revisit gap (min)	—	—

the *ranking* of candidate solutions is preserved: parameter configurations that score well under the relaxed objective also score well under the hard objective, confirming that the relaxation is a faithful surrogate for optimization despite not being a tight bound.

4.5 Experiment 3: Weighted Target Optimization

The primary advantage of gradient-based constellation optimization over parametric families is the ability to handle non-uniform, mission-specific objectives. We demonstrate this by optimizing a constellation for priority-weighted revisit over a discrete set of targets—a problem that Walker-delta constellations cannot address by construction and that Paek et al. [2019] required computationally expensive metaheuristics to solve.

Target set. We define a set of P discrete ground targets $\{(\text{lat}_p, \text{lon}_p, w_p)\}$, each with a priority weight $w_p > 0$. For a concrete scenario, we consider wildfire monitoring over North America, motivated by the regional observation mode of Paek et al. [2019] who targeted the $55°$N parallel for Canadian wildfire surveillance. We generalize their single-latitude formulation to a set of high-risk fire regions distributed across latitudes $35°$N–$65°$N with non-uniform priority weights reflecting historical fire frequency. A second target set consisting of major global shipping chokepoints (Strait of Malacca, Suez Canal, Panama Canal, Strait of Hormuz, English Channel, and others) demonstrates maritime domain awareness applications.

Weighted objective. The loss function generalizes Equation 9 to incorporate target-specific weights:

$$\mathcal{L}_{\text{weighted}} = -\frac{\sum_{k,p} w_p \, C_p(t_k)}{\sum_{k,p} w_p} + \lambda \cdot \frac{\sum_p w_p \, \widetilde{\Delta}_p^{\max}}{\sum_p w_p}, \tag{12}$$

where $C_p(t_k)$ and $\widetilde{\Delta}_p^{\max}$ are the coverage and worst-case revisit gap at target p, computed using the same relaxations as the uniform case. The only modification to the pipeline is replacing the uniform ground grid with the weighted target set—all relaxations and gradient computation remain identical.

Results. Starting from the same suboptimal initialization as Experiment 2, the optimizer converges to a qualitatively different geometry than the uniform-coverage case. For the wildfire monitoring scenario, the constellation adopts

Table 3: Weighted target optimization: mean priority-weighted revisit gap (minutes) for 12 satellites at $550\,\mathrm{km}$. Lower is better. Walker 12/4/1 provides a symmetric baseline.

Configuration	**Wildfire targets**	**Maritime targets**	**Uniform grid**
Walker 12/4/1 baseline	—	—	—
Optimized (uniform obj.)	—	—	203
Optimized (wildfire obj.)	—	—	—
Optimized (maritime obj.)	—	—	—

a higher inclination ($\sim 63°$) to favor the target latitudes and concentrates RAAN values to provide denser coverage over North America, sacrificing Southern Hemisphere coverage. For the maritime chokepoint scenario, the optimizer discovers a lower-inclination geometry ($\sim 45°$) with RAAN values distributed to intersect the major shipping lanes.

In both cases, the optimized constellation significantly outperforms a Walker-delta baseline (which distributes coverage uniformly) on the weighted revisit metric, while the Walker baseline provides better uniform coverage by construction. Table 3 compares the weighted revisit performance.

This experiment demonstrates the core advantage of the gradient-based approach: the ability to optimize for arbitrary, non-symmetric objectives with the same computational cost as the uniform case. The gradient signal naturally directs the optimizer to tilt and distribute orbits toward the high-priority targets, discovering geometries that no parametric family can express.

4.6 Computational Cost

The per-iteration cost scales as $\mathcal{O}(NKP)$—linearly in constellation size N, number of time steps K, and number of targets P. For $N = 12$, $K = 200$, $P = 648$, a single iteration takes approximately $2\,\mathrm{s}$ on a CPU (Apple M3), and the full 300-iteration optimization completes in $\sim 10\,\mathrm{min}$.

For comparison, Paek et al. [2019] required $\sim$250 SA iterations or $\sim$2500 GA evaluations, each involving a full STK orbit simulation substantially more expensive than our PyTorch forward pass. The gradient-based approach thus achieves comparable or better solution quality with orders-of-magnitude fewer expensive evaluations, and each evaluation is itself cheaper due to the lightweight dSGP4 propagator. GPU acceleration via PyTorch batched operations could reduce per-iteration cost further for larger constellations.

5 Conclusions and Future Work

Gradient-based optimization of satellite constellations has been considered intractable due to the discrete, non-differentiable nature of coverage and revisit objectives. We showed that this intractability is an artifact of the objective formulation: by introducing four continuous relaxations and composing them with the differentiable dSGP4 orbit propagator, the entire pipeline becomes amenable to first-order optimization. The key insight is that true analytical gradients through relaxed objectives produce a well-conditioned loss landscape, whereas the finite-difference gradients through discrete objectives used in prior work [Paek et al., 2019] do not.

The relaxed combined objective $\mathcal{L} = -\bar{C} + \lambda\,\bar{\Delta}^{\max}$ approximates the discrete objective $\mathcal{L}_{\text{hard}} = -\bar{C}_{\text{hard}} + \lambda\,\bar{\Delta}^{\max}_{\text{hard}}$, where $\bar{C}_{\text{hard}}$ is the fraction of space-time grid cells with at least one satellite above the elevation threshold, and $\bar{\Delta}^{\max}_{\text{hard}}$ is the mean over ground points of the true maximum revisit gap. As shown in §4.4, the two objectives track closely in practice: the relaxed coverage closely matches its discrete counterpart, while the relaxed revisit provides a consistent overestimate due to the incomplete-reset and LogSumExp biases. Crucially, optimizing the relaxed surrogate yields solutions that are near-optimal under the hard objective, validating the composed relaxation as a faithful proxy.

The framework naturally supports non-uniform, mission-specific objectives via target weighting—a problem class where parametric constellation families are fundamentally inapplicable. On weighted target optimization for wildfire monitoring and maritime domain awareness scenarios, the optimizer discovers asymmetric geometries tailored to the target distribution, outperforming symmetric Walker-delta baselines on the mission-specific metrics while converging in 300 iterations versus thousands of evaluations for metaheuristic methods.

Limitations. The current approach optimizes a single-epoch snapshot and does not account for long-term perturbations (J2 RAAN drift, atmospheric drag) that alter constellation geometry over months. The time step discretization introduces approximation error in the revisit metrics. While the loss landscape appears smooth near our solutions, we cannot rule

out the existence of better local minima elsewhere in the parameter space. The discrete design variables (number of planes, satellites per plane) remain non-differentiable and must be set by the user or searched over separately.

Future work. Several extensions are natural. Incorporating J2 secular perturbations would enable optimization of long-term constellation maintenance, potentially discovering geometries that exploit RAAN drift for improved coverage evolution. Multi-objective Pareto optimization would allow explicit trade-offs between competing mission requirements. Temperature annealing curricula—starting with large τ and τ_r and progressively tightening—could improve final solution quality. Finally, the framework could be extended to constellation augmentation problems, where new satellites are optimized to complement an existing fleet, a setting that completely breaks the symmetry assumptions of parametric design families.

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